MSG No. 218.83 $P\bar{4}'3n'$ [Type III, cubic]

Table 1: Wyckoff site: `2a`, site symmetry: `23.`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`

Table 2: Wyckoff site: `6b`, site symmetry: `222..`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,8,10,12]`
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[6,7,9,11]`
4	$\left[\frac{1}{2}, 0, 0\right]$	`[13,14,21,22]`
5	$\left[0, 0, \frac{1}{2}\right]$	`[15,16,23,24]`
6	$\left[0, \frac{1}{2}, 0\right]$	`[17,18,19,20]`

Table 3: Wyckoff site: `6c`, site symmetry: `-4'..`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[1,2,13,14]`
2	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[3,4,21,22]`
3	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	`[5,12,17,19]`
4	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[6,9,16,23]`
5	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[7,11,15,24]`
6	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[8,10,18,20]`

Table 4: Wyckoff site: `6d`, site symmetry: `-4'..`

No.	position	mapping
1	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[1,2,13,14]`
2	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[3,4,21,22]`
3	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	`[5,12,17,19]`
4	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[6,9,16,23]`
5	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[7,11,15,24]`
6	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	`[8,10,18,20]`

Table 5: Wyckoff site: `8e`, site symmetry: `.3.`

No.	position	mapping
1	$[x, x, x]$	`[1,5,6]`
2	$[x, -x, -x]$	`[2,10,11]`
3	$[-x, x, -x]$	`[3,7,12]`
4	$[-x, -x, x]$	`[4,8,9]`

continued ...

Table 5

No.	position	mapping
5	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$	[13,18,23]
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$	[14,15,19]
7	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	[16,17,21]
8	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	[20,22,24]

Table 6: Wyckoff site: 12f, site symmetry: 2..

No.	position	mapping
1	$[x,\ 0,\ 0]$	[1,2]
2	$[-x,\ 0,\ 0]$	[3,4]
3	$[0,\ x,\ 0]$	[5,12]
4	$[0,\ 0,\ x]$	[6,9]
5	$[0,\ 0,\ -x]$	[7,11]
6	$[0,\ -x,\ 0]$	[8,10]
7	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[13,14]
8	$\left[\frac{1}{2},\ \frac{1}{2},\ x+\frac{1}{2}\right]$	[15,24]
9	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-x\right]$	[16,23]
10	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[17,19]
11	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[18,20]
12	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[21,22]

Table 7: Wyckoff site: 12g, site symmetry: 2..

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ 0\right]$	[1,2]
2	$\left[-x,\ \frac{1}{2},\ 0\right]$	[3,4]
3	$\left[0,\ x,\ \frac{1}{2}\right]$	[5,12]
4	$\left[\frac{1}{2},\ 0,\ x\right]$	[6,9]
5	$\left[\frac{1}{2},\ 0,\ -x\right]$	[7,11]
6	$\left[0,\ -x,\ \frac{1}{2}\right]$	[8,10]
7	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	[13,14]
8	$\left[\frac{1}{2},\ 0,\ x+\frac{1}{2}\right]$	[15,24]
9	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-x\right]$	[16,23]
10	$\left[0,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[17,19]
11	$\left[0,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[18,20]
12	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[21,22]

Table 8: Wyckoff site: 12h, site symmetry: 2..

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{2}\right]$	[1,2]
2	$\left[-x,\ 0,\ \frac{1}{2}\right]$	[3,4]
3	$\left[\frac{1}{2},\ x,\ 0\right]$	[5,12]

continued ...

Table 8

No.	position	mapping
4	$\left[0,\ \frac{1}{2},\ x\right]$	[6,9]
5	$\left[0,\ \frac{1}{2},\ -x\right]$	[7,11]
6	$\left[\frac{1}{2},\ -x,\ 0\right]$	[8,10]
7	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	[13,14]
8	$\left[0,\ \frac{1}{2},\ x+\frac{1}{2}\right]$	[15,24]
9	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-x\right]$	[16,23]
10	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	[17,19]
11	$\left[\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	[18,20]
12	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[21,22]

Table 9: Wyckoff site: 24i, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[x,\ -y,\ -z]$	[2]
3	$[-x,\ y,\ -z]$	[3]
4	$[-x,\ -y,\ z]$	[4]
5	$[z,\ x,\ y]$	[5]
6	$[y,\ z,\ x]$	[6]
7	$[-y,\ z,\ -x]$	[7]
8	$[-z,\ -x,\ y]$	[8]
9	$[-y,\ -z,\ x]$	[9]
10	$[z,\ -x,\ -y]$	[10]
11	$[y,\ -z,\ -x]$	[11]
12	$[-z,\ x,\ -y]$	[12]
13	$\left[\frac{1}{2}-x,\ z+\frac{1}{2},\ \frac{1}{2}-y\right]$	[13]
14	$\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ y+\frac{1}{2}\right]$	[14]
15	$\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ x+\frac{1}{2}\right]$	[15]
16	$\left[z+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-x\right]$	[16]
17	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[17]
18	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[18]
19	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[19]
20	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[20]
21	$\left[x+\frac{1}{2},\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$	[21]
22	$\left[x+\frac{1}{2},\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$	[22]
23	$\left[\frac{1}{2}-z,\ y+\frac{1}{2},\ \frac{1}{2}-x\right]$	[23]
24	$\left[z+\frac{1}{2},\ y+\frac{1}{2},\ x+\frac{1}{2}\right]$	[24]